

Gebze Technical University

Dep. of Mechanical Engineering

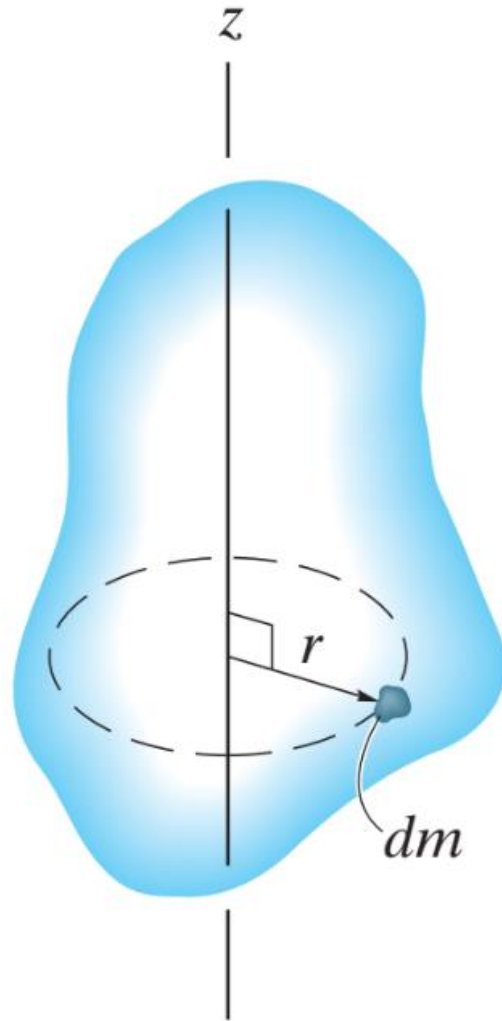
Statics (ME 231)

Lecture 19:

Mass Moments of Inertia

By Saeed Lotfan

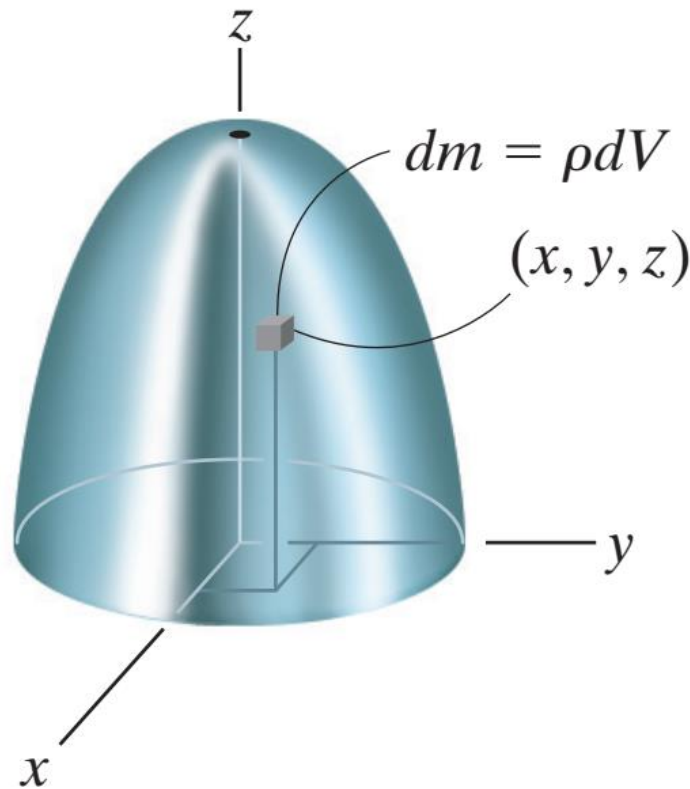
Mass Moment of Inertia



$$I = \int_m r^2 dm$$

Mass Moment of Inertia

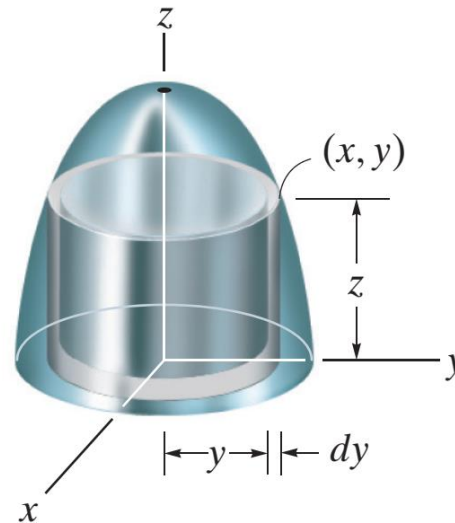
If a body is symmetrical with respect to an axis, then its mass moment of inertia about the axis can be determined by using a single integration. Shell and disk elements are used for this purpose.



Mass Moment of Inertia

Shell Element

- If a *shell element* having a height z , radius y , and thickness dy is chosen for integration, then its volume is $dV = (2\pi y)(z) dy$.

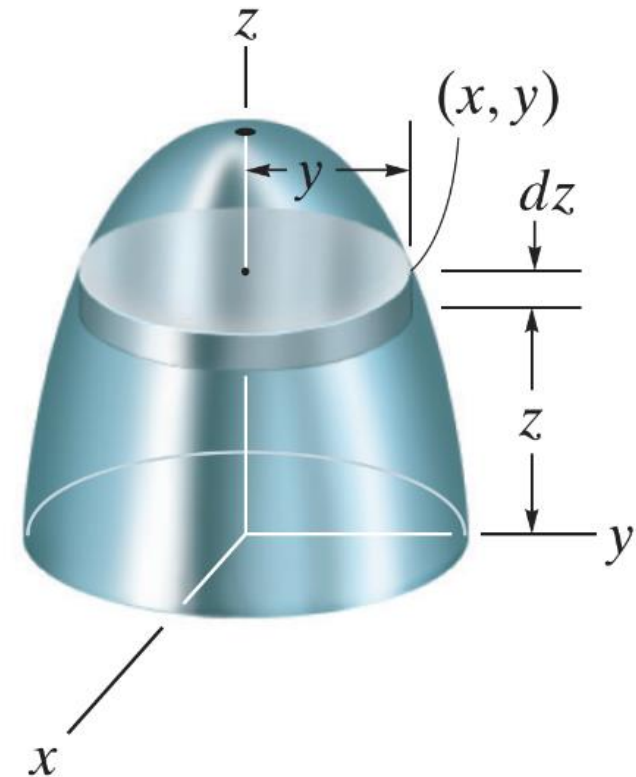


- This element can be used for determining the moment of inertia I_z since the *entire element*, due to its “thinness,” lies at the *same* perpendicular distance $r = y$ from the z axis.

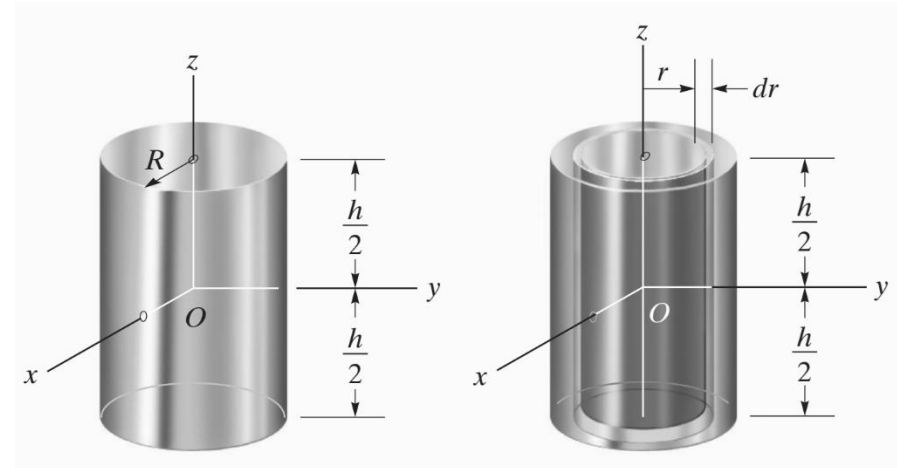
Mass Moment of Inertia

Disk Element

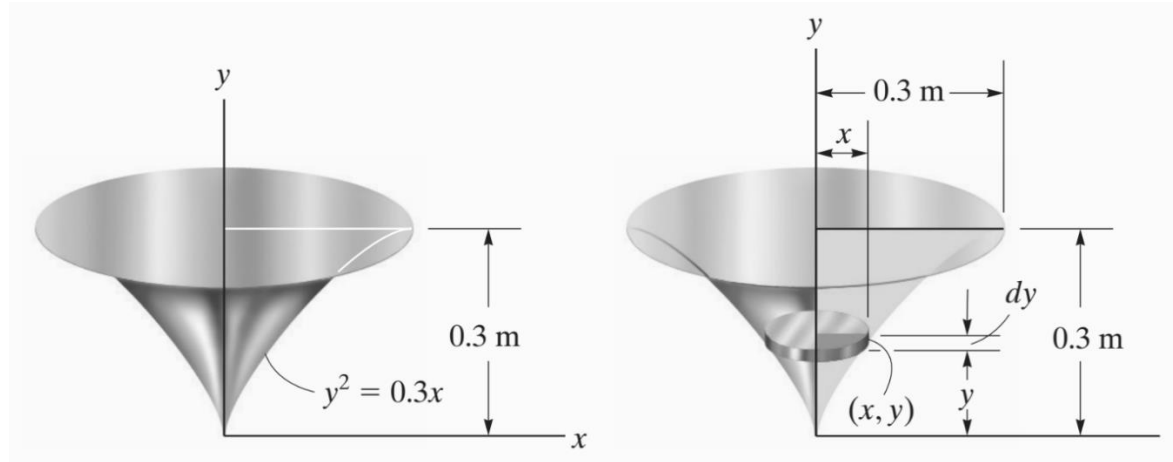
- If a disk element having a radius y and a thickness dz is chosen for integration, then its volume is $dV = (\pi y^2) dz$.
- In this case the element is *finite* in the radial direction, and consequently its points *do not* all lie at the *same radial distance* r from the z axis. As a result, basic definitions *cannot* be used to determine I_z . Instead, to perform the integration using this element, it is first necessary to determine the moment of inertia *of the element* about the z axis and then integrate this result.



Example: Determine the mass moment of inertia of the cylinder shown below about the z axis. The density of the material is ρ .



Example: Determine the mass moment of inertia of the cylinder shown below about the y axis. The density of the material is $\rho = 2.5 \text{ Mg/m}^3$.



Mass Moment of Inertia

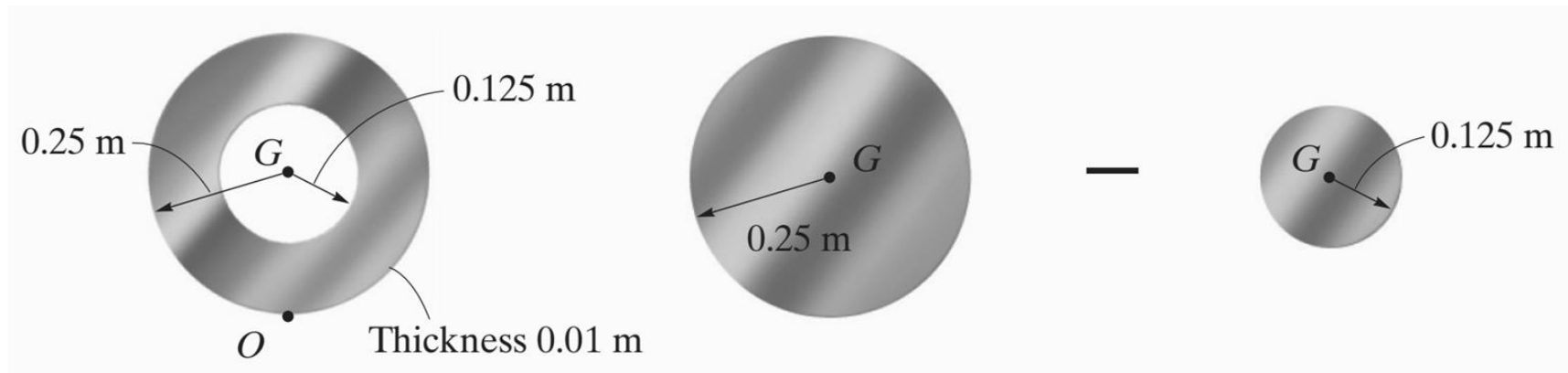
Parallel-Axis Theorem

$$I = I_G + md^2$$

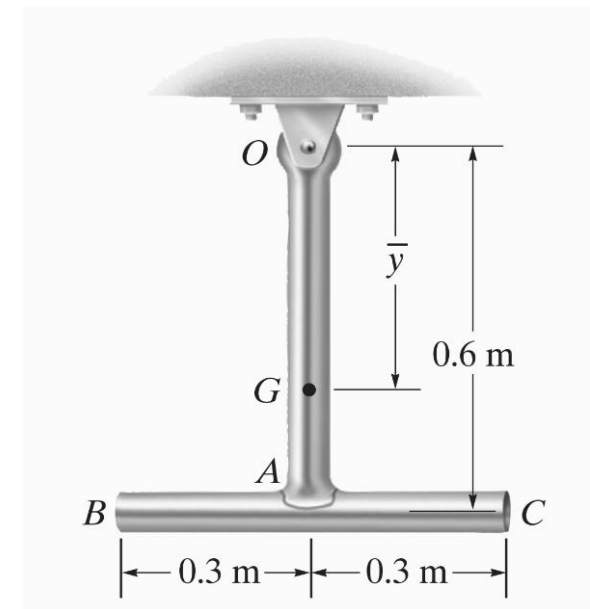
Radius of Gyration

$$I = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I}{m}}$$

Example: If the plate shown below has a density of 8000 kg/m^3 and a thickness of 10 mm , determine its mass moment of inertia about an axis perpendicular to the page and passing through point O .

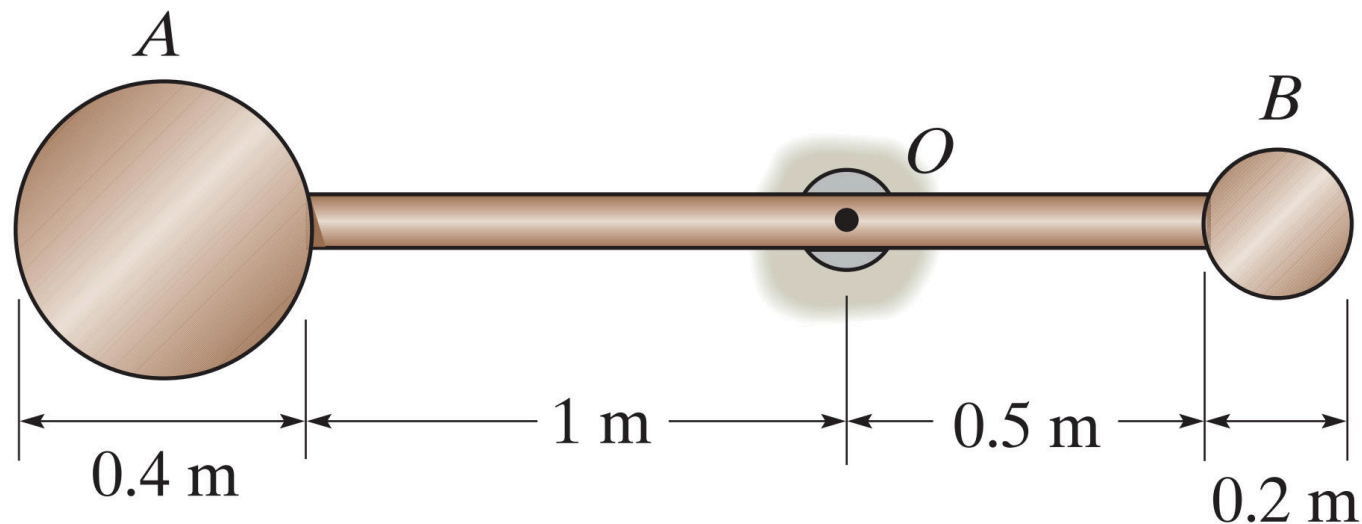


Example: The pendulum below consists of two thin rods each having a mass of 5 kg. Determine the pendulum's mass moment of inertia about an axis passing through (a) the pin at O , and (b) the mass center G of the pendulum.



Practice at Home

1. The pendulum consists of an 8-kg circular disk A , a 2-kg circular disk B , and a 4-kg slender rod. Determine the radius of gyration of the pendulum about an axis perpendicular to the page and passing through point O .



Practice at Home

2. The pendulum consists of two slender rods AB and OC which have a mass of 3 kg/m . The thin plate has a mass of 12 kg/m^2 . Determine the location $\bar{y}$ of the center of mass G of the pendulum, then calculate the moment of inertia of the pendulum about an axis perpendicular to the page and passing through G .

